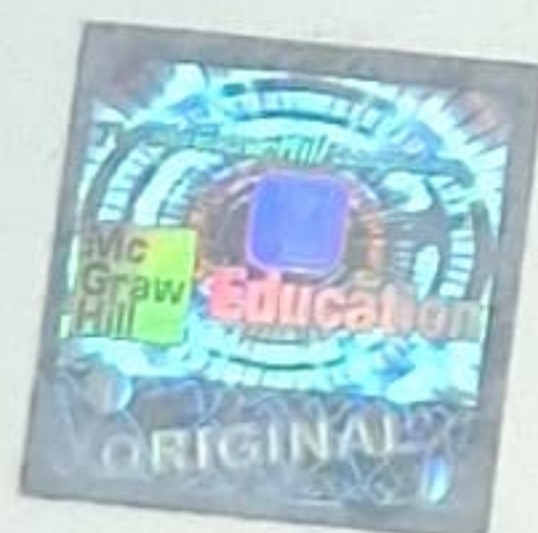


JNTU

Electronic Circuit Analysis



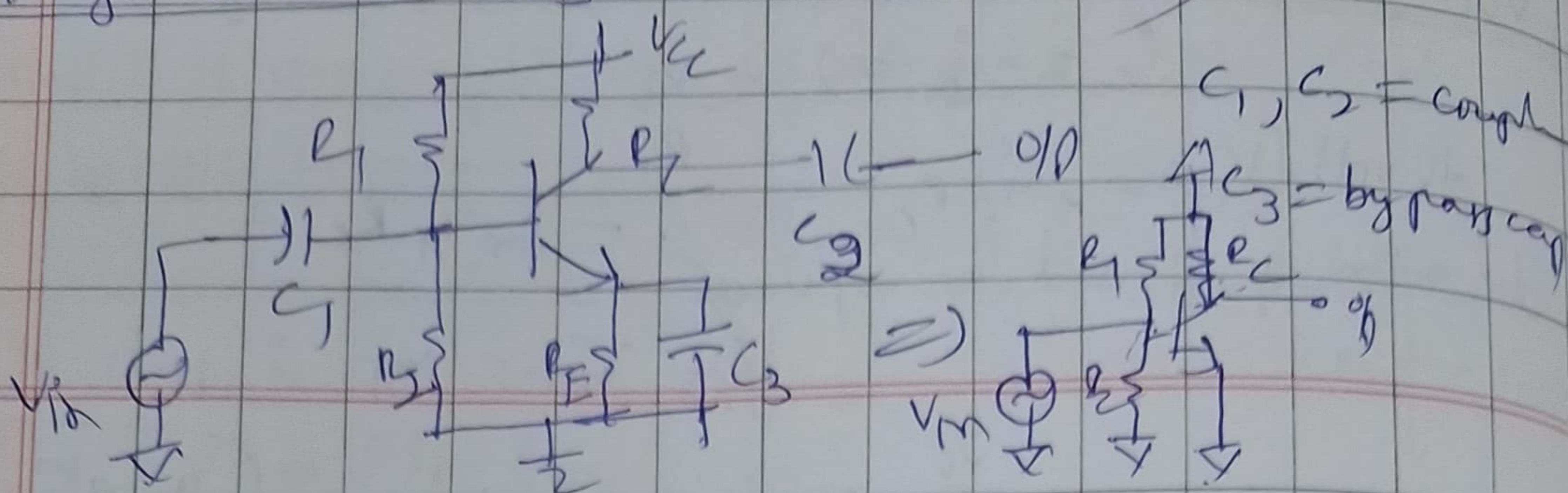
S Salivahanan
N Suresh Kumar

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Hyderabad

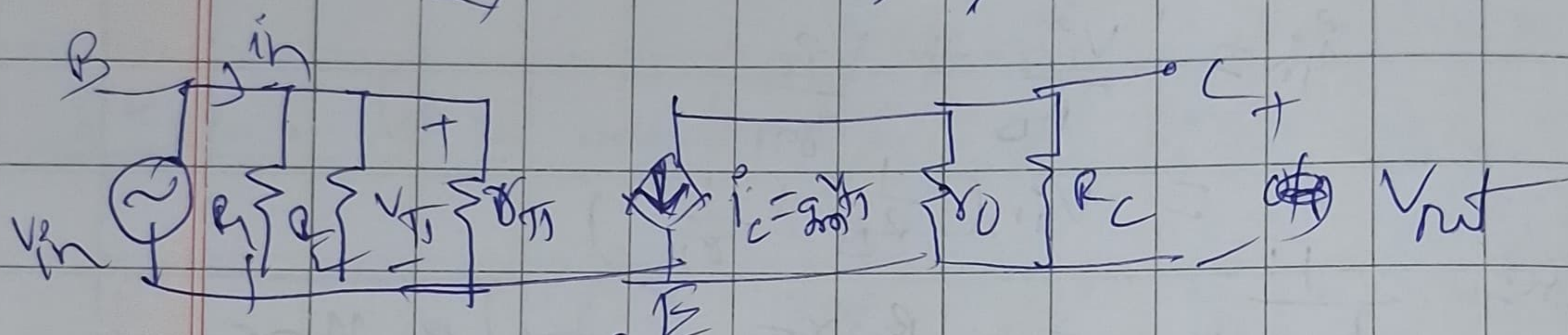
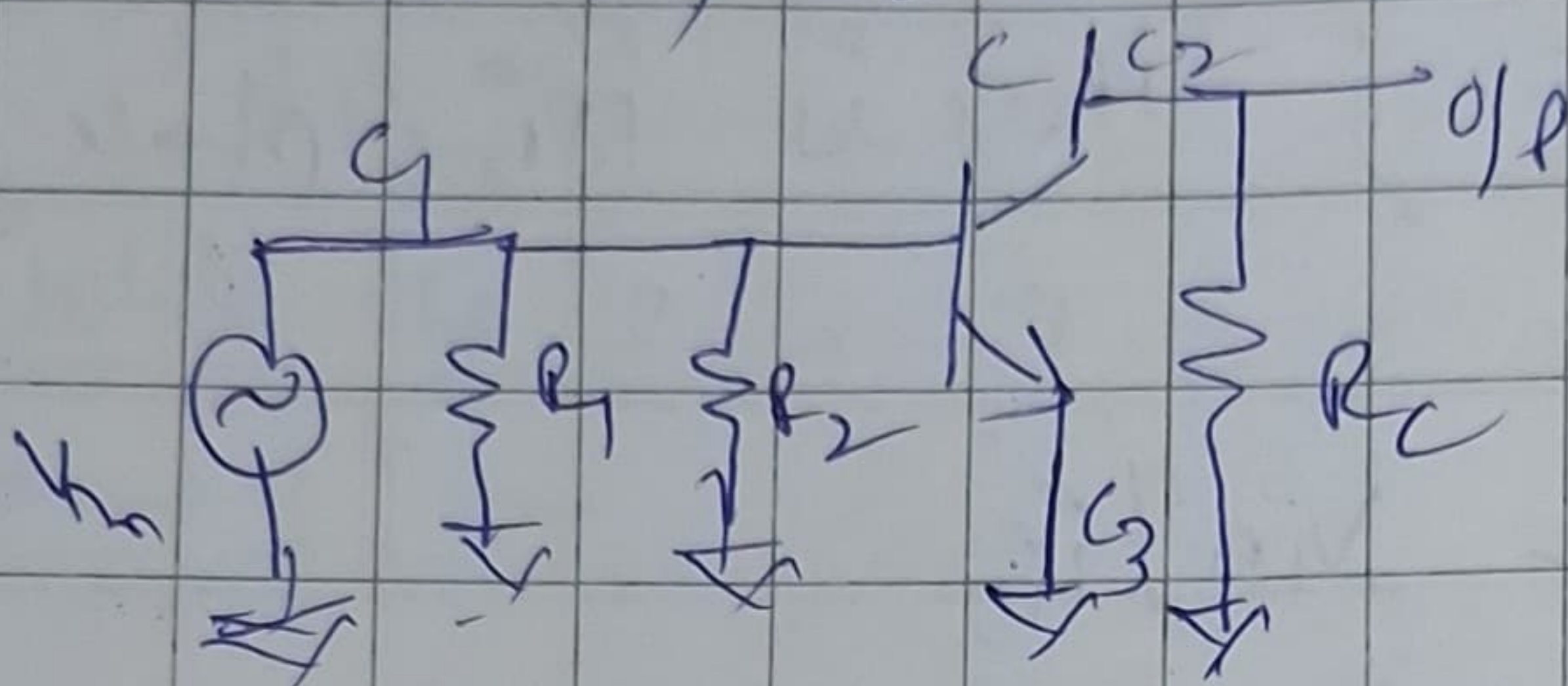


Image to PDF

Voltage divider bias (Small Signal Analysis)



C_3 is short circuited; R_E also short circuited



$$V_{out} = -i_c (r_o \parallel R_C) = -g_m V_{\pi} (r_o \parallel R_C)$$

$$A_v = \frac{V_o}{V_{in}} = -g_m (r_o \parallel R_C) \quad \begin{matrix} \nearrow V_{in} \\ 180^\circ \text{ PS b/w } V_{in} \text{ and } V_{out} \end{matrix}$$

$$Z_{in} = \frac{V_{in}}{I_{in}} = \text{applied KCL}$$

$$I_{in} = \frac{V_{in}}{R_1} + \frac{V_{in}}{R_2} + \frac{V_{in}}{r_{\pi}}$$

$$\frac{V_{in}}{I_{in}} = Z_{in} = R_1 \parallel R_2 \parallel r_{\pi}$$

$$(R_1 \parallel R_2) \gg r_{\pi}$$

$$Z_{in} = R_1 \parallel R_2$$

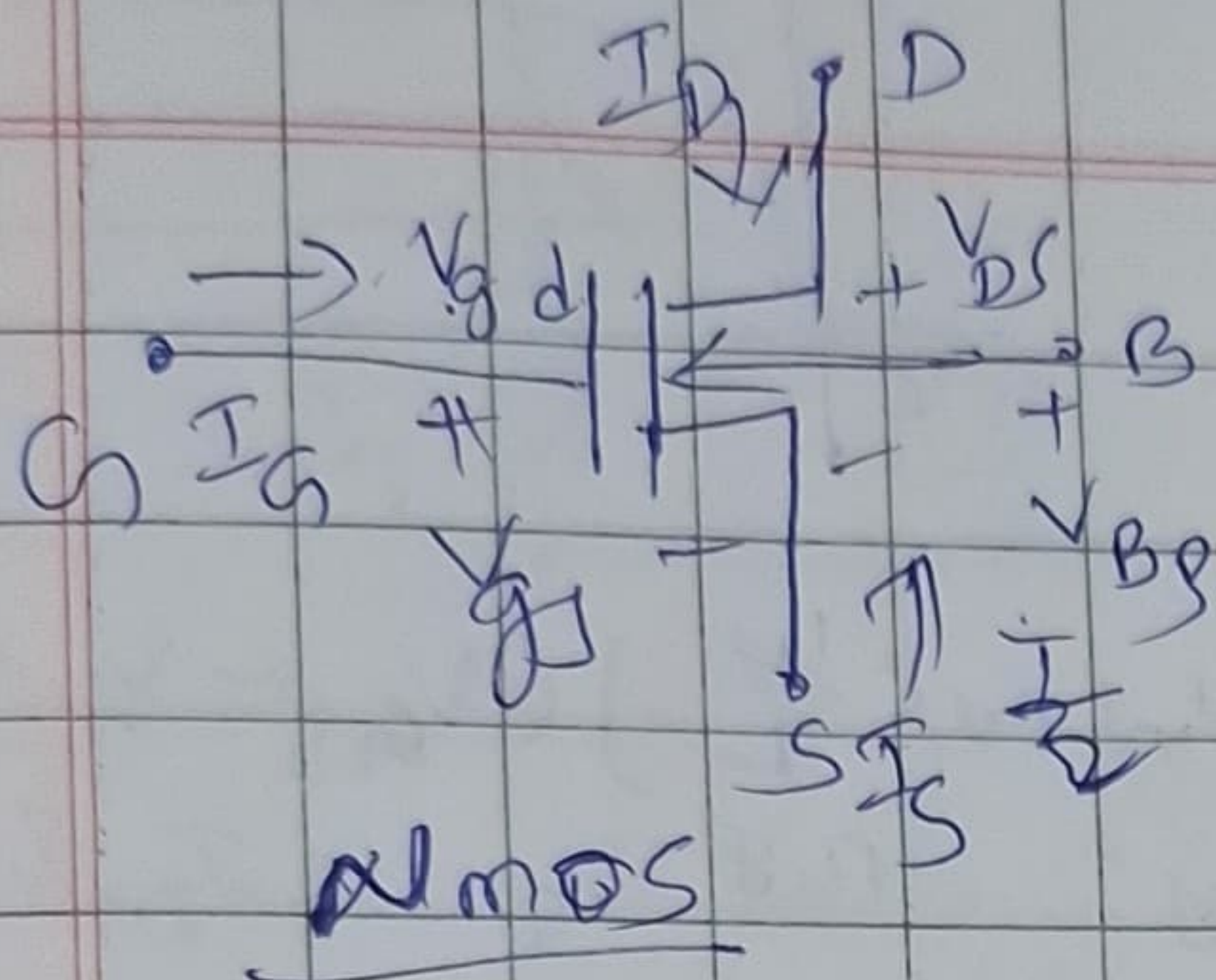
o/r I_{in} :

$$R_{TH} = R_G \parallel R_D$$

$$Z_0 = R_D \parallel R_C$$

MOSFET large signal (DC model): low freq

gate controlled C.S.

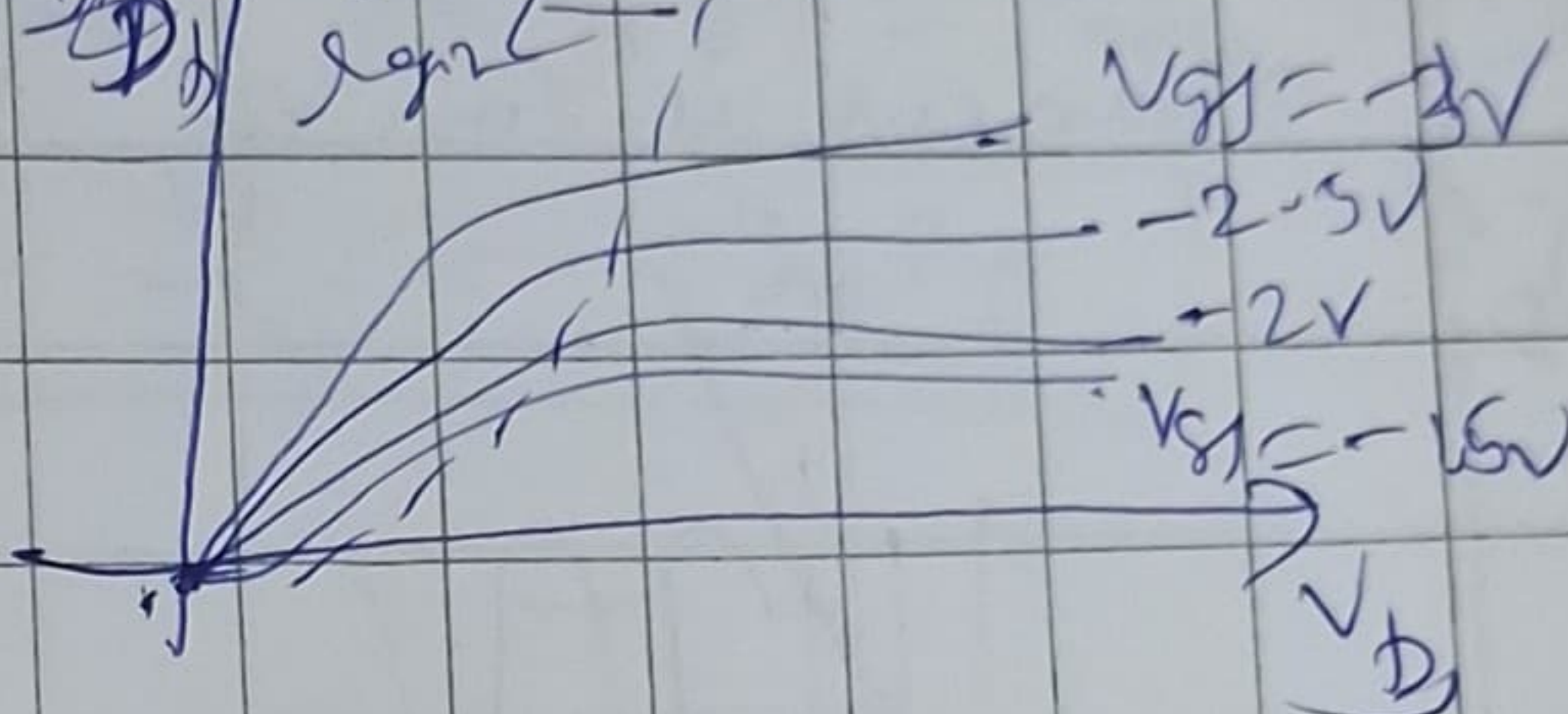


$L \rightarrow$ chnl length

$w \rightarrow$ width of chnl

NMOS

ohmic region / saturation



$$V_{DS} = V_{GS} - V_T = V_P$$

$$V_{DS} > V_P \Rightarrow \text{sat.}$$

$$I_D = \begin{cases} \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{GS} - V_T)^2 V_{DS} & ; V_{GS} > V_T \\ 0 & ; V_{GS} < V_T \end{cases}$$

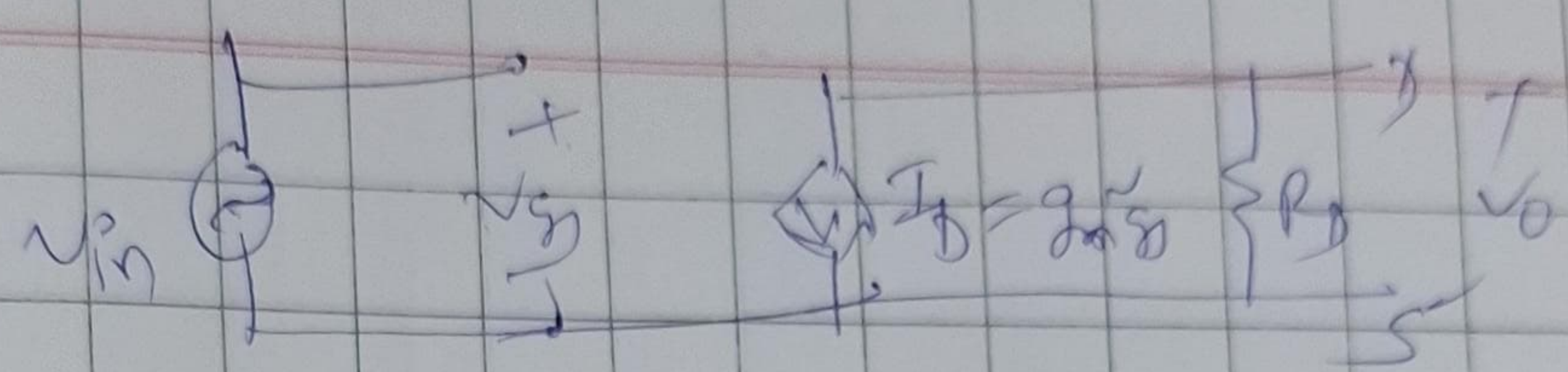
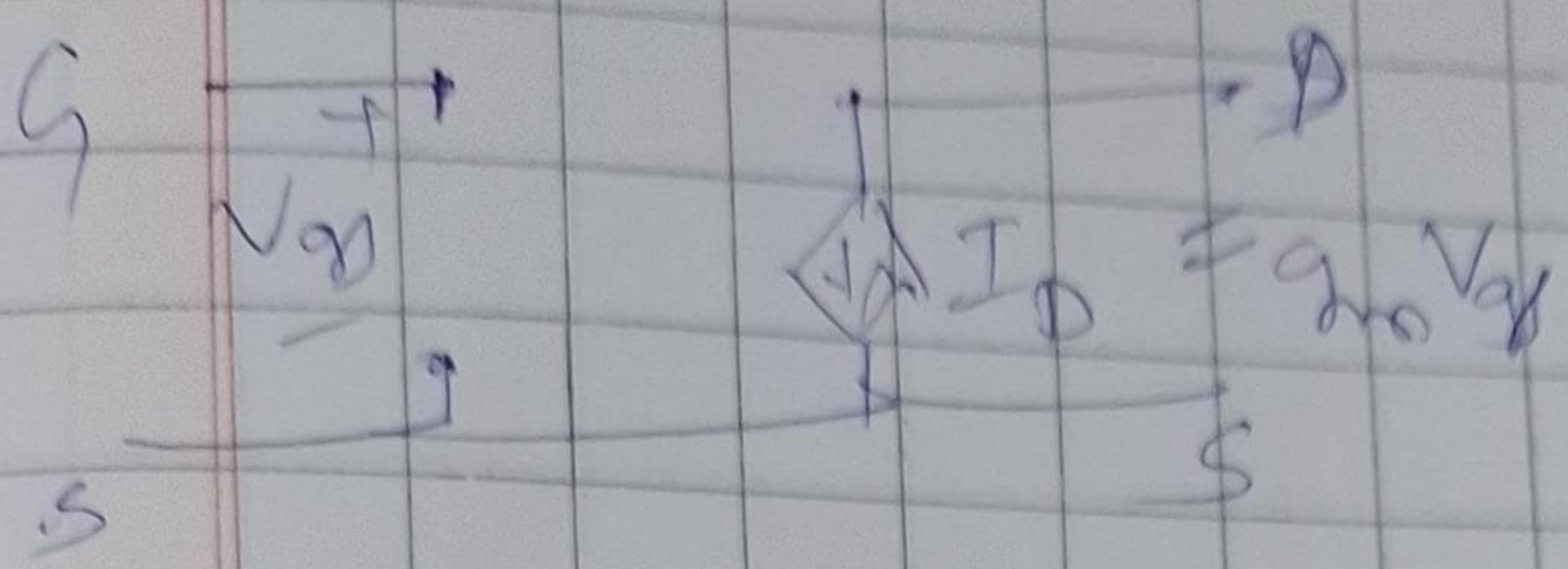
$$I_D = 0 ; V_{GS} < V_T \text{ (cut-off)}, V_{DS} \geq 0$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{GS} - V_T - \frac{V_{DS}}{2}) V_{DS} ; V_{GS} > V_T \text{ (ohmic)}$$

$$0 < V_{DS} < V_{GS} - V_T$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{GS} - V_T)^2 ; V_{GS} > V_T ; V_{DS} > V_{GS} - V_T \text{ (sat)}$$

The four hybrid parameters
 $h_{11} = \frac{V_1}{I_1}$
 $h_{12} = \frac{V_1}{V_2}$
 $h_{21} = \frac{I_2}{I_1}$
 $h_{22} = \frac{I_2}{V_2}$



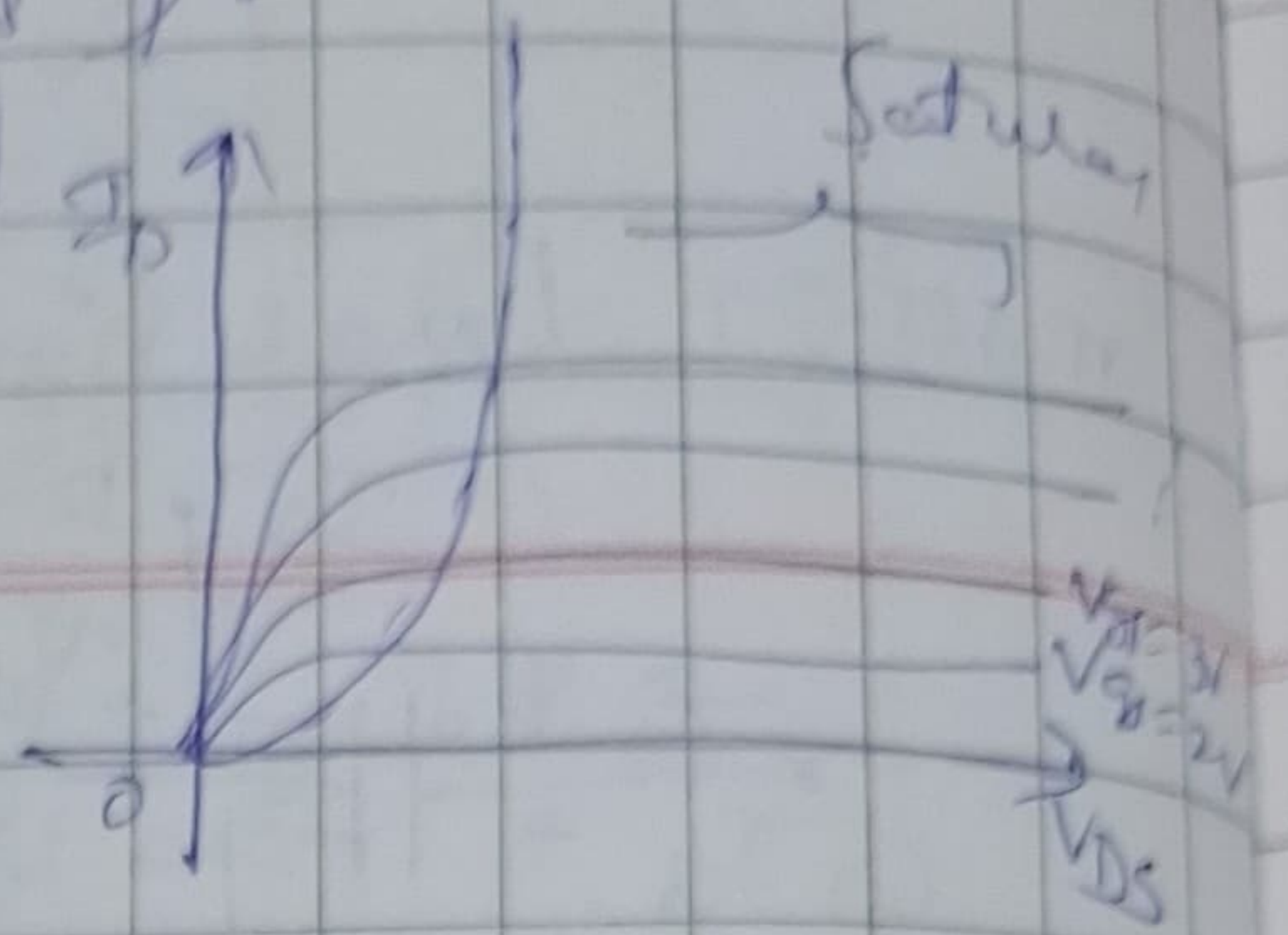
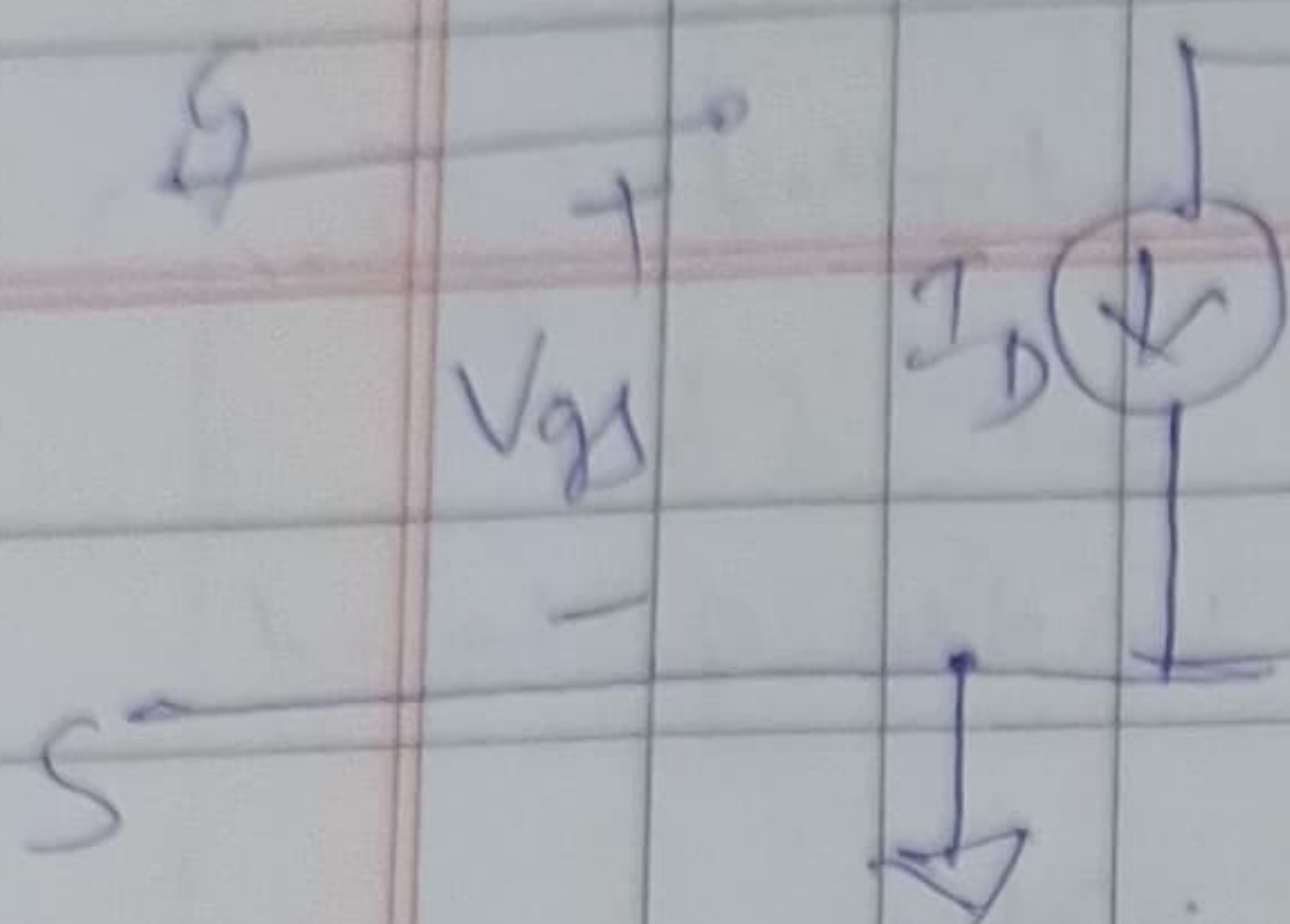
$$V_o = -g_m V_{gs} R_D = -g_m V_{in} R_D$$

$$\boxed{A_v = \frac{V_o}{V_i} = -g_m R_D}$$

Mosfet Small signal model :

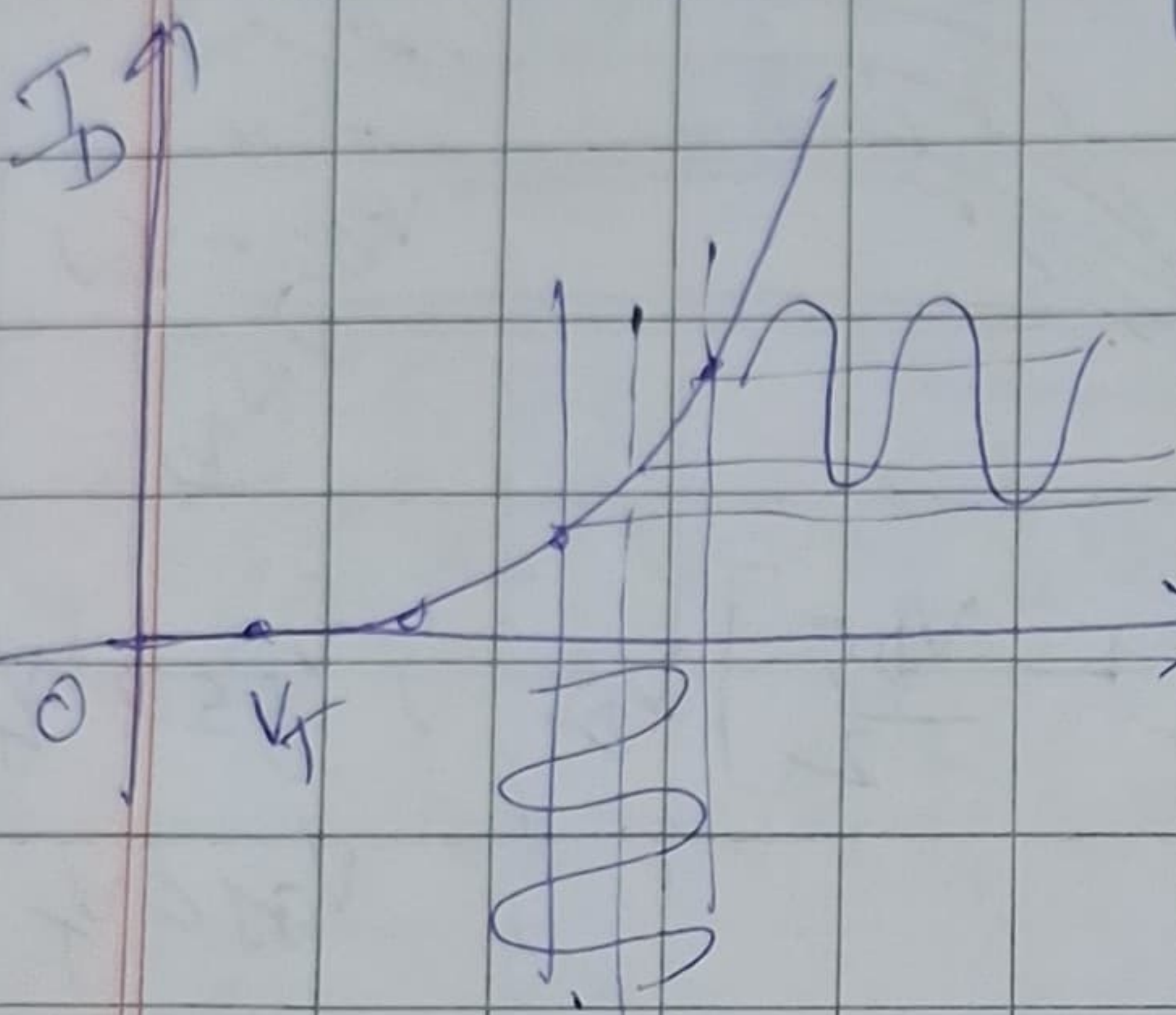
In sat region mosfet acts as μr
 Inductive region (V_{DS} cont c.s)

Drain char



$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_T)^2$$

Drain cur is fun of Vgs. But Vgs vs ID is non-linear

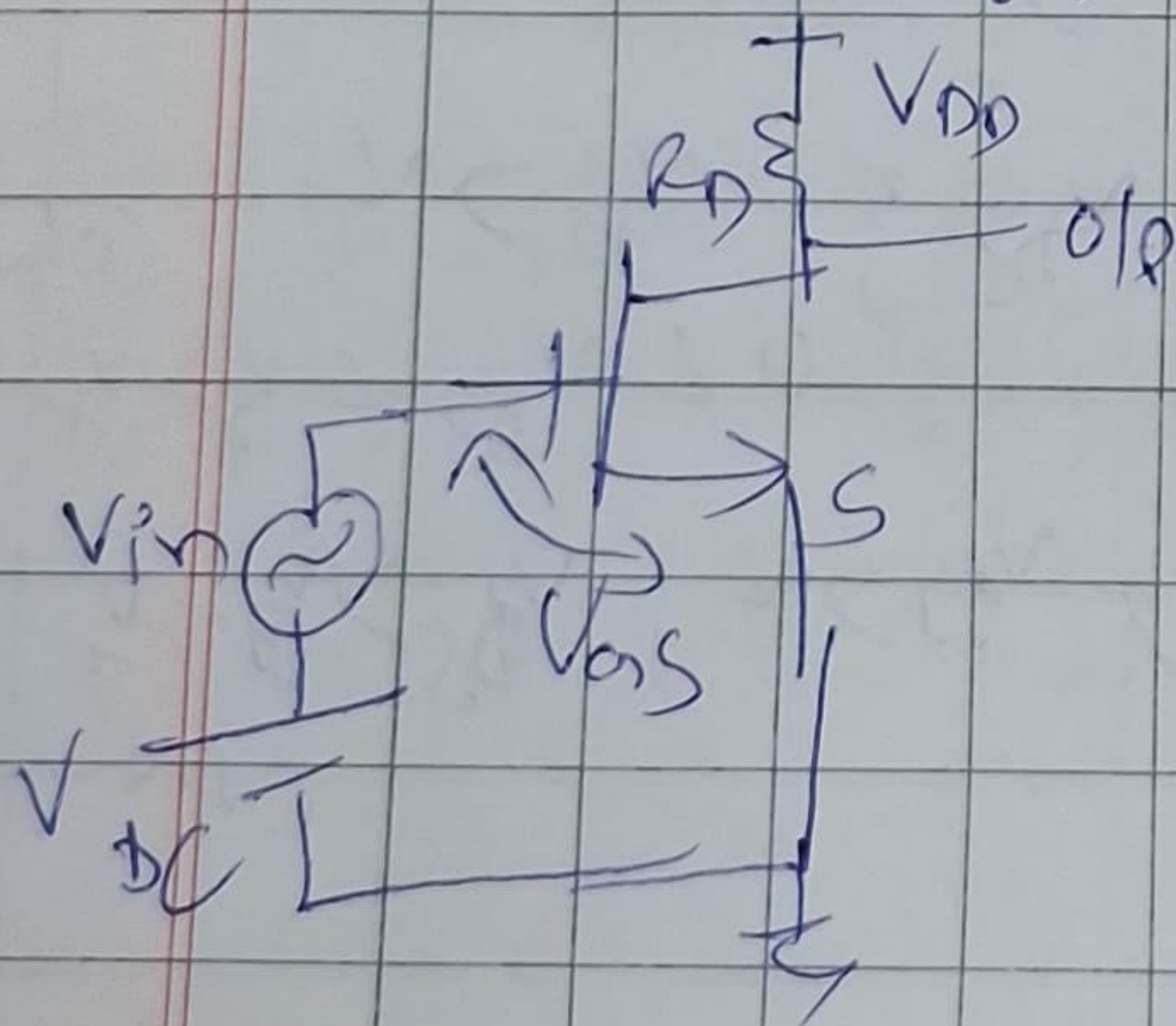


Amount of charge in Vgs
 is change ID. V is
 bias dependent.

trans on = $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$
 it is slope of curve at op pt

$$g_m \propto \frac{W}{L}$$

$$g_m = \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_T)$$



$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_T)^2$$

$$V_{GS} = V_{DS} + V_{i^2}$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{i^2} + V_{DS} - V_T)^2$$

simplified $V_{i^2} \ll V_{DS} - V_T$

$$I_D = \underbrace{I_{D0}}_{DC} + \underbrace{g_m V_{in}}_{AC}$$

Example 3.5 Calculate the coupling capacitor C_C required in Fig. 3.17 to provide a low frequency 3 dB point at 125 Hz if $R_S = 600 \Omega$, $h_{ie} = 1 \text{ k}\Omega$, $h_{fe} = 60$, $R_1 = 5 \text{ k}\Omega$ and $R_2 = 1.25 \text{ k}\Omega$. For (a) an ideal bypass capacitor C_E , (b) a practical bypass capacitor with $R_{CE} = 25 \Omega$.

Solution:

The lower 3 dB frequency, $f_L = \frac{1}{2\pi(R_S + R'_i)C_C}$

$$(a) \quad R'_i = R_1 \parallel R_2 \parallel h_{ie}$$

$$C_C = \frac{1}{2\pi f_L (R_S + R'_i)} = \frac{1}{125 \times 2\pi \times [600 + 5000 \parallel 1.25 \times 10^3 \parallel 1000]} \\ = \frac{1}{125 \times 2\pi \times [600 + 500]} \\ C_C = 1.15 \mu\text{F}$$

$$(b) \quad R'_i = R_1 \parallel R_2 \parallel [h_{ie} + (1 + h_{fe}) R_{CE}] \\ = 5000 \parallel 1.25 \times 10^3 \parallel [1000 + (61)(25)] = 716.31 \Omega$$

$$C_C = \frac{1}{2\pi f_L (R_S + R'_i)} \\ = \frac{1}{125 \times 2\pi \times [600 + 716.31]} = 0.97 \mu\text{F}$$

3.8 HIGH FREQUENCY HYBRID π MODEL FOR COMMON EMITTER TRANSISTOR MODEL

At high frequencies, the capacitive effects of the transistor junctions and the delay in response of the transistor caused by the process of diffusion of carriers should be taken into account in determining the high frequency model of a transistor.

A high frequency- π or Giacoletto model for a transistor is shown in Fig. 3.20. $r_{bb'}$ —base spreading resistance between the actual base B and virtual base B' . It represents the bulk resistance of the base. Its typical value is 100Ω . $r_{b'e}$ —resistance between the virtual base B' and the emitter terminal E . Typical value is $1 \text{ k}\Omega$.

Input resistance from base to emitter with the output shorted is simply $r_{bb'} + r_{b'e}$ and this is the same as h_{ie} . Hence, $h_{ie} = r_{bb'} + r_{b'e}$.

$r_{b'c}$ —resistance between the virtual base B' and the collector terminal C . It has a large value (typical value = $4 \text{ M}\Omega$).

$C_{b'e}$ —diffusion capacitance of the normally forward biased base-emitter junction. It has a typical value of 100 pF .

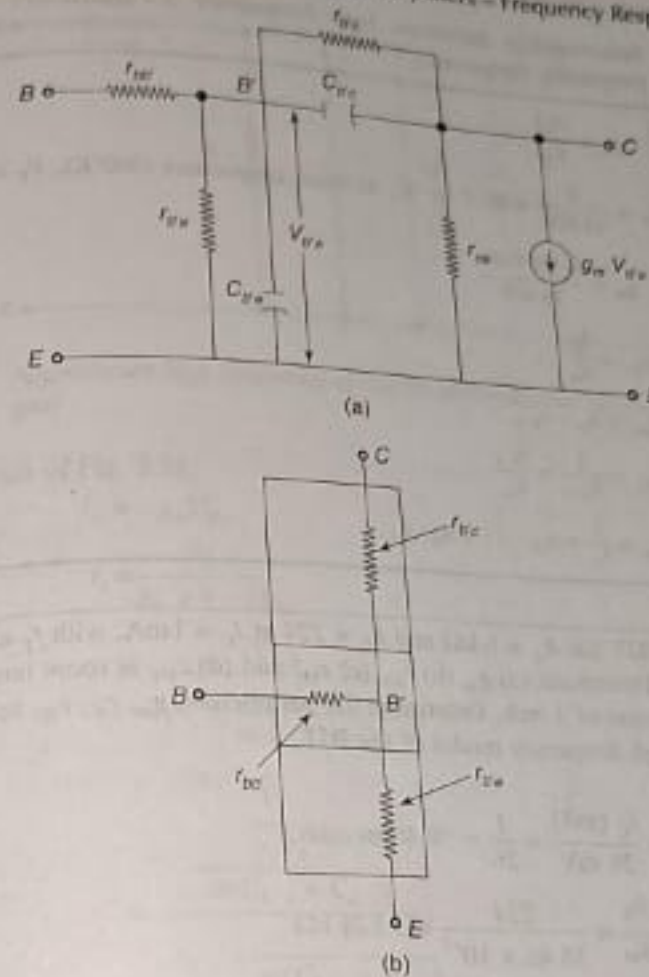


Fig. 3.20 (a) Hybrid π model for a transistor in the CE connection; (b) Diagram showing virtual base B' and ohmic base-spreading resistance $r_{bb'}$.

$C_{b'c}$ —transistor capacitance of the normally reverse biased collector-base junction. It has a typical value of 3 pF .

r_{ce} —output resistance with a typical value of $80 \text{ k}\Omega$. Since $r_{ce} \gg R_L$, if a load R_L is connected r_{ce} can be neglected.

$g_m V_{b'e}$ —output current generator value where g_m is the transconductance of the transistor.

For a given collector current the conductances and resistances of the hybrid π circuit can be calculated from the low frequency h -parameter values from the equations given in Table 3.2.

Table 3.2 Relationship between Low Frequency h-Parameters and High Frequency Parameters

(i)	$g_m = \frac{I_C}{V_T}$
	where $V_T = \frac{T}{11,600}$ with T in °K. At room temperature (300°K), $V_T \approx 0.026$ V
	so that $g_m = \frac{I_C \text{ (in mA)}}{26 \text{ mV}}$
(ii)	$r_{b'e} = \frac{h_{ie}}{g_m}$
(iii)	$r_{bb'} = h_{ie} - r_{b'e}$
(iv)	$r_{b'e} = \frac{1}{g_m} = \frac{r_{b'e}}{h_{fe}}$
(v)	$g_{ce} = \frac{1}{r_{ce}} = h_{oe} - (1 + h_{fe}) g_{b'e}$

Example 3.6 A BJT has $h_{ie} = 6 \text{ k}\Omega$ and $h_{fe} = 224$ at $I_C = 1 \text{ mA}$, with $f_T = 80 \text{ MHz}$ and $C_{b'e} = 12 \text{ pF}$. Determine (a) g_m (b) $r_{b'e}$ (c) $r_{bb'}$ and (d) $C_{b'e}$ at room temperature and a collector current of 1 mA. Determine the parameters, g_m , $r_{b'e}$, $r_{bb'}$ and $C_{b'e}$ of the small signal high frequency model of the BJT.

Solution:

$$g_m = \frac{I_C \text{ (mA)}}{26 \text{ mV}} = \frac{1}{26} = 38.46 \text{ m mho}$$

$$r_{b'e} = \frac{h_{ie}}{g_m} = \frac{224}{38.46 \times 10^{-3}} = 5.824 \text{ k}\Omega$$

$$r_{bb'} = h_{ie} - r_{b'e} = 6000 - 5824 = 176 \Omega$$

$$\begin{aligned} C_{b'e} &= \frac{g_m}{2\pi f_T} - C_{b'c} \\ &= \frac{38.46 \times 10^{-3}}{2\pi \times 80 \times 10^6} - 12 \times 10^{-12} \\ &= 76.5 \times 10^{-12} - 12 \times 10^{-12} = 64.5 \text{ pF} \end{aligned}$$

3.8.1 CE Short-circuit Current Gain

The transistor's high frequency capability can be known, if its CE short-circuit forward-current transfer ratio or current gain is found as a function of frequency.

As R_L is short circuited the approximate high frequency model becomes as shown in Fig. 3.21, where $C_{b'e}$ is parallel with $C_{b'c}$.

The short circuit current gain is given by

$$A_i = \frac{I_L}{I_i}$$

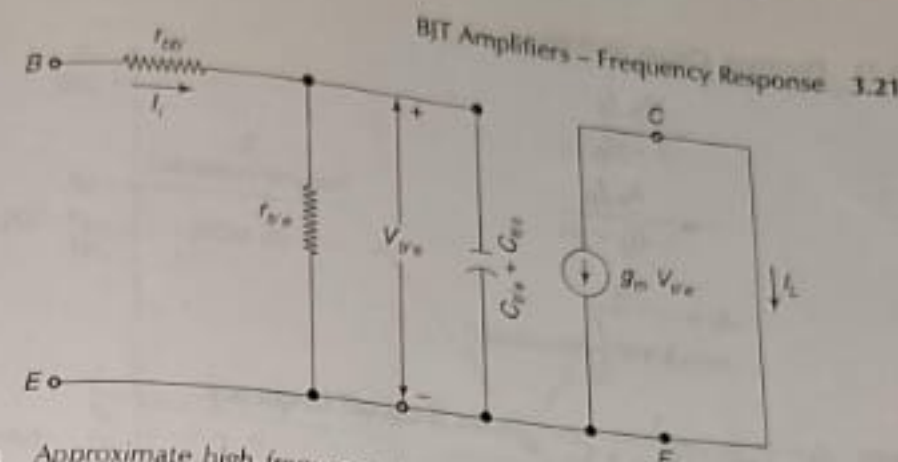


Fig. 3.21 Approximate high frequency circuit for determination of short circuit current gain

From circuit of Fig. 3.21,

$$I_L = -g_m V_{be}$$

$$I_i = \frac{V_{be}}{r_{b'e} \parallel -jX_{C_{b'e}}}$$

$$= \frac{V_{be}}{r_{b'e} \parallel \frac{-j}{\omega(C_{b'e} + C_{b'c})}}$$

$$= \frac{V_{be}}{\frac{-jr_{b'e}}{\omega(C_{b'e} + C_{b'c})}}$$

$$= \frac{V_{be}}{-j(r_{b'e})^2 f \beta} \left[\text{Let } f \beta = \frac{1}{2\pi r_{b'e} (C_{b'e} + C_{b'c})} \right]$$

$$= \frac{V_{be}}{-jr_{b'e} f \beta}$$

$$\text{Current gain, } A_i = \frac{-g_m V_{be}}{\frac{-jr_{b'e} f \beta}{f - jf \beta}} = \frac{jg_m r_{b'e} f \beta}{f - jf \beta}$$

$$\begin{aligned}
 &= \frac{h_{fe}}{r_{b'e}} \times \frac{1}{2\pi(C_{b'e} + C_{b'c})} \\
 &= \frac{g_m}{2\pi(C_{b'e} + C_{b'c})}
 \end{aligned} \quad (3.45)$$

Since $C_{b'c} \gg C_{b'e}$

$$f_T = \frac{g_m}{2\pi C_{b'e}} \quad \text{or} \quad C_{b'e} = \frac{g_m}{2\pi f_T}$$

Substituting $f_\beta = \frac{f_T}{h_{fe}}$ in Eq. (3.36)

$$A_i = \frac{-h_{fe}}{1 + jh_{fe}\left(\frac{f}{f_T}\right)} \quad (3.46)$$

This equation shows the dependence of transistor's short circuit current gain on the transistor's gain at low frequencies h_{fe} and the high frequency characteristics f_T .

Example 3.7 A BJT has $g_m = 38$ mhos, $r_{b'e} = 5.9$ k Ω , $h_{fe} = 6$ k Ω , $r_{bb'} = 100$ Ω , $C_{b'e} = 12$ pF, $C_{b'c} = 63$ pF, and $h_{fe} = 224$ at 1 kHz. Calculate α and β cut-off frequencies and f_T .

Solution:

$$\begin{aligned}
 f_\alpha &= \frac{h_{fe}}{2\pi r_{b'e} C_{b'e}} \\
 &= \frac{224}{2\pi \times 5.9 \times 10^3 \times 63 \times 10^{-12}} = 95.9 \text{ MHz}
 \end{aligned}$$

$$\begin{aligned}
 f_\beta &= \frac{1}{2\pi r_{b'e} (C_{b'e} + C_{b'c})} \\
 &= \frac{1}{2\pi \times 5.9 \times 10^3 (63 \times 10^{-12} + 12 \times 10^{-12})} = 0.359 \text{ MHz}
 \end{aligned}$$

$$\begin{aligned}
 f_T &= \frac{g_m}{2\pi (C_{b'e} + C_{b'c})} \\
 &= \frac{38 \times 10^{-3}}{2\pi (63 \times 10^{-12} + 12 \times 10^{-12})} = 80.63 \text{ MHz}
 \end{aligned}$$

3.8.5 High Frequency Current Gain with Resistive Load

With a resistive load connected in the output, the high frequency equivalent circuit of a CE transistor amplifier is shown in Fig. 3.23.

By using Miller's theorem the circuit of Fig. 3.23 can be modified as described below.

From Fig. 3.23, the voltage gain,

$$A = \frac{V_{CE}}{V_{b'e}} = \frac{-g_m V_{b'e} R_L}{V_{b'e}} = -g_m R_L$$

$$1 - A = 1 - (-g_m R_L) = 1 + g_m R_L$$

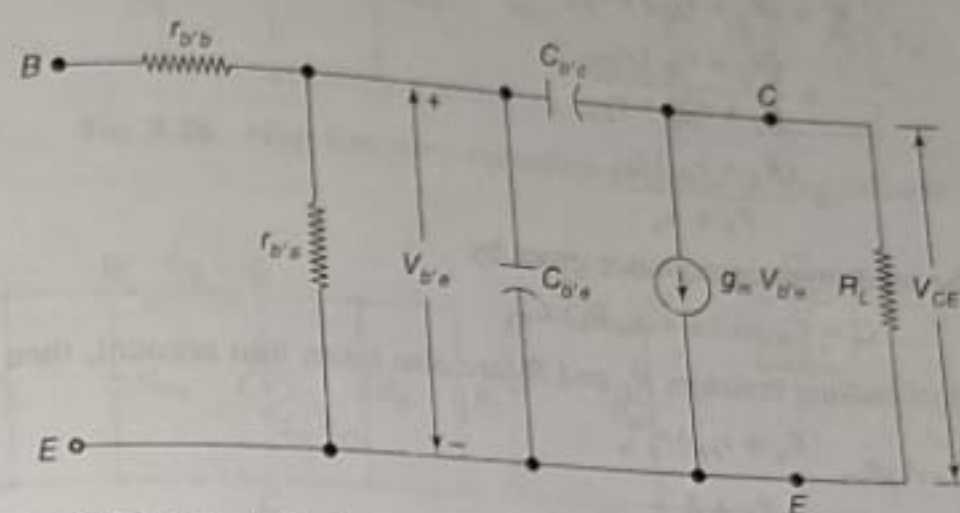


Fig. 3.23 High frequency equivalent circuit with resistive load

Since the impedance at the input gets decreased by a factor of $(1 - A)$ the capacitance will be increased by a factor of $(1 - A)$ or $1 + g_m R_L$.

The capacitance that is to be included in the output circuit will not make any significant change in the performance and may be neglected. This results in the modified equivalent circuit of Fig. 3.24.

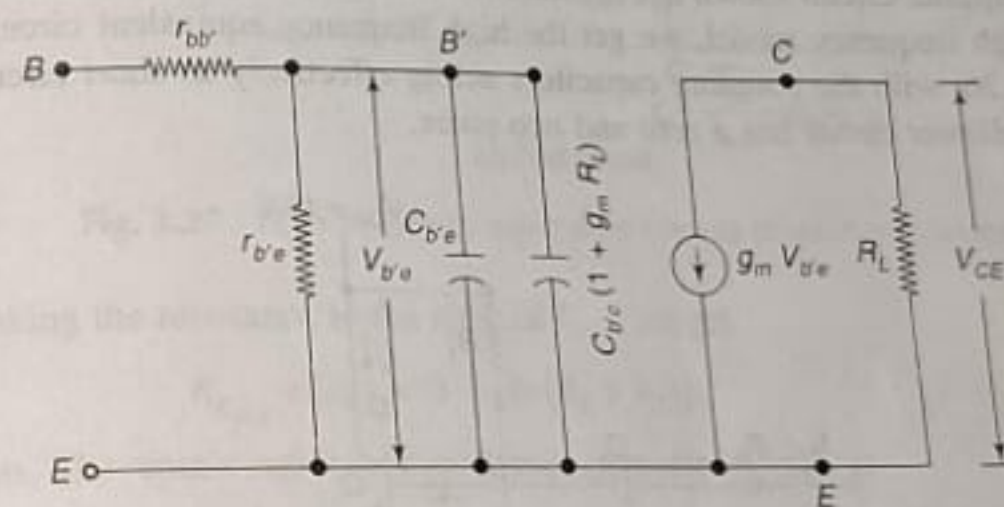


Fig. 3.24 Modified equivalent circuit

The total input capacitance between B' and E is

$$C = C_{b'e} + (1 + g_m R_L) C_{b'c} \quad (3.47)$$

Since the circuit of Fig. 3.24 is essentially the same as that of Fig. 3.21, Eq. (3.37) can be used to find the upper 3 dB frequency. Hence, the upper dB frequency,

$$f_2 = \frac{1}{2\pi r_{b'e} C} \quad (3.48)$$

As C has an increased value, f_2 is appreciably lower than that calculated for f_β .
Effect of source resistance on frequency response: If the effect of source resistance

$$\begin{aligned}
 &= \frac{\beta h_{fe} f_{\beta}}{f - j f_{\beta}} \\
 &= \frac{h_{fe} f_{\beta}}{-j(f + j f_{\beta})} \\
 &= \frac{-h_{fe}}{1 + j \frac{f}{f_{\beta}}}
 \end{aligned}
 \quad (3.36)$$

where $f_{\beta} = \frac{1}{2\pi r_{b'e} (C_{b'e} + C_{b'c})}$ is the β cut-off frequency. (3.37)

3.8.2 β Cut-Off Frequency

The β cut-off frequency f_{β} , also referred to as f_{hfe} , is the CE short-circuit small signal forward-current-transfer-ratio cut off frequency.

f_{β} is the frequency at which a transistor's CE short-circuit current gain drops 3-dB from its value at lower (mid) frequencies. f_{β} represents the maximum attainable bandwidth for the current gain of a CE amplifier with a given transistor.

3.8.3 α Cut-Off Frequency

A CB amplifier has a much higher 3 dB frequency than a CE amplifier. The short-circuit current gain for CB amplifier which can be derived from the approximate high frequency circuit of the CB amplifier with output shorted is given by

$$A_i = \frac{I_L}{I_i} = \frac{-h_{fb}}{1 + j \left(\frac{f}{f_{\alpha}} \right)} \quad (3.38)$$

$$\text{where } f_{\alpha} = \frac{1}{2\pi r_{b'e} (1 + h_{fb}) C_{b'e}} = \frac{h_{fe}}{2\pi r_{b'e} C_{b'e}} \quad (3.39)$$

Substituting Eq. (3.37) in Eq. (3.39),

$$f_{\alpha} = \frac{h_{fe} f_{\beta} (C_{b'e} + C_{b'c})}{C_{b'e}} \quad (3.40)$$

f_{α} is the α (alpha) cut-off frequency at which the CB short-circuit small signal forward-current transfer ratio (A_i) drops 3 dB from its value at low frequencies (≈ 1 kHz). Figure 3.22 shows the variation of A_i with frequency for CE and CB amplifiers and f_{α} and f_{β}

3.8.4 Gain-Bandwidth Product

f_{α} and f_{β} defined in the last section gives an idea about the high-frequency capability of a transistor. An even more important characteristic is f_T , which is defined as the frequency at which the short-circuit common-emitter current gain has a magnitude of unity.

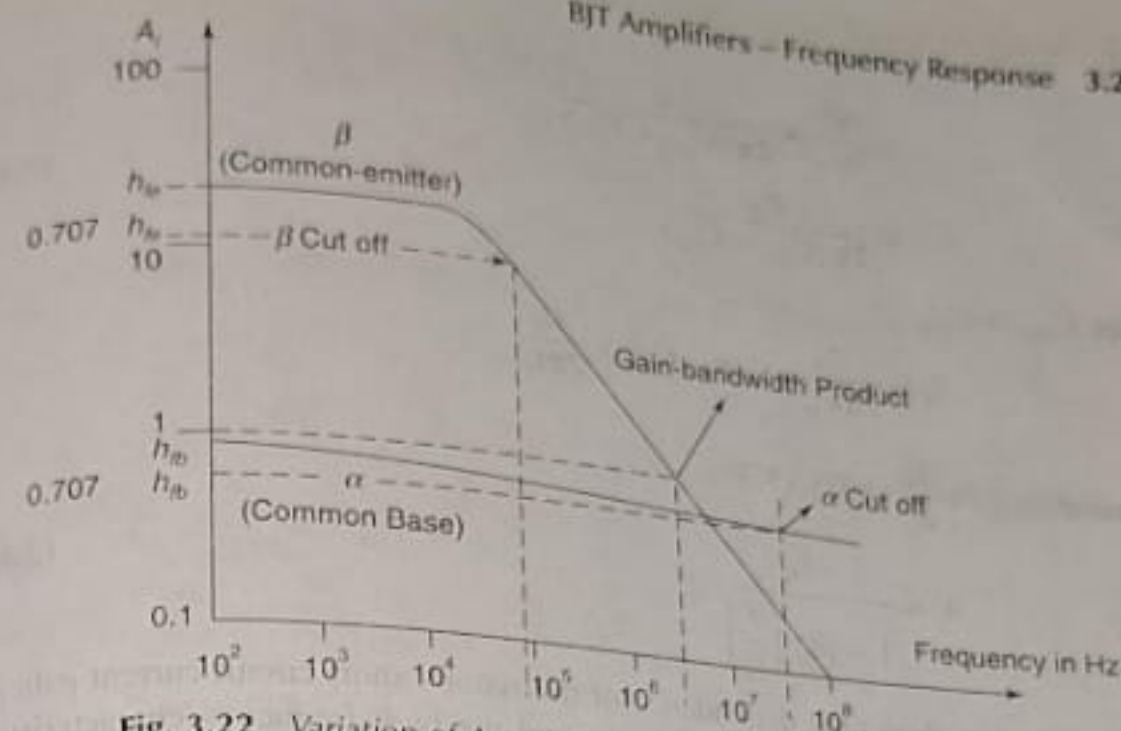


Fig. 3.22 Variation of A_i with frequency for CE and CB amplifiers

$$\text{From Eq. (3.36), } |A_i| = \frac{h_{fe}}{\sqrt{1 + (f/f_{\beta})^2}} \quad (3.41)$$

and at $f = f_T$,

$$|A_i| = 1$$

From Fig. 3.22 f_T is less than f_{α} but much greater than f_{β} and $\left(\frac{f_T}{f_{\beta}} \right)^2 \gg 1$

Hence, Eq. (3.41) reduces to

$$1 = \frac{h_{fe}}{\frac{f_T}{f_{\beta}}}$$

Then

$$\frac{f_T}{f_{\beta}} = h_{fe} \quad (3.42)$$

$$f_T = h_{fe} f_{\beta} \quad (3.43)$$

f_T is the product of low frequency current gain h_{fe} and β cut-off frequency, f_{β} , or CE bandwidth. Similarly, it can be proved that

$$f_T = h_{fb} f_{\alpha} \quad (3.44)$$

Value of f_T range from 1 MHz for audio transistor up to 1 GHz for high frequency transistors. Typical values of f_{β} , f_{α} and f_T are 0.36 MHz, 95.9 MHz and 80.63 MHz.

Substituting for f_{β} from Eq. (3.37) in (3.43),

$$f_T = h_{fe} \frac{1}{2\pi r_{b'e} (C_{b'e} + C_{b'c})}$$

R_S is also taken into account the upper 3 dB frequency f_2 is given by

$$f_2 = \frac{1}{2\pi R' C} \quad (3.49)$$

where

$$\begin{aligned} R' &= (R_S + r_{bb'}) \parallel r_{b'e} \\ &= \frac{(R_S + r_{bb'}) r_{b'e}}{R_S + r_{bb'} + r_{b'e}} \\ &= \frac{(R_S + r_{bb'}) r_{b'e}}{R_S + h_{ie}} \end{aligned}$$

and C is the total input capacitance given by

$$C = C_{b'e} + (1 + g_m R_L) C_{b'c}$$

If the effect of biasing resistors R_1 and R_2 are also taken into account, then

$$R' = \frac{(R'_S + r_{bb'}) r_{b'e}}{R'_S + h_{ie}} \quad (3.50)$$

where

$$R'_S = R_S \parallel R_B \quad \text{and} \quad R_B = R_1 \parallel R_2$$

Thus, the source and biasing resistors have a strong influence in determining the upper 3 dB frequency f_2 .

3.9 EMITTER FOLLOWER AT HIGHER FREQUENCIES

Let us analyse the high frequency response of an emitter follower i.e., common collector amplifier circuit shown in Fig. 3.25. If the transistor of the circuit is replaced by its high frequency model, we get the high frequency equivalent circuit shown in Fig. 3.26, with the coupling capacitors acting effectively as short circuits. The emitter follower circuit has a zero and two poles.

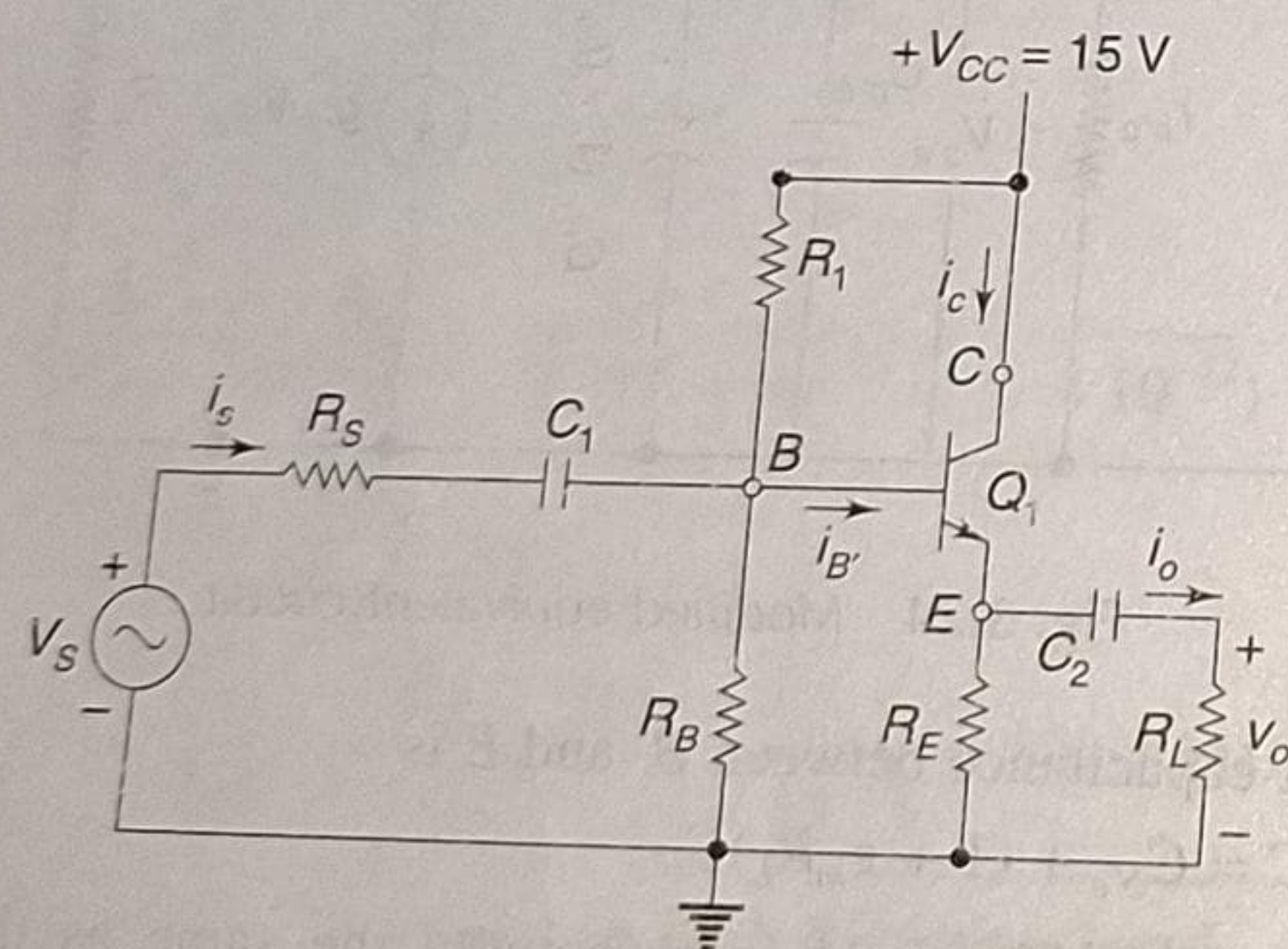


Fig. 3.25 Emitter follower circuit

Assuming that $C_{b'e}$ is open-circuited and $V_S = 0$, the equivalent circuit is as shown in Fig. 3.27(a). Looking the resistance to the left of $C_{b'c}$, we get

$$R_{C_{b'e1}} = R_B \parallel R_S$$

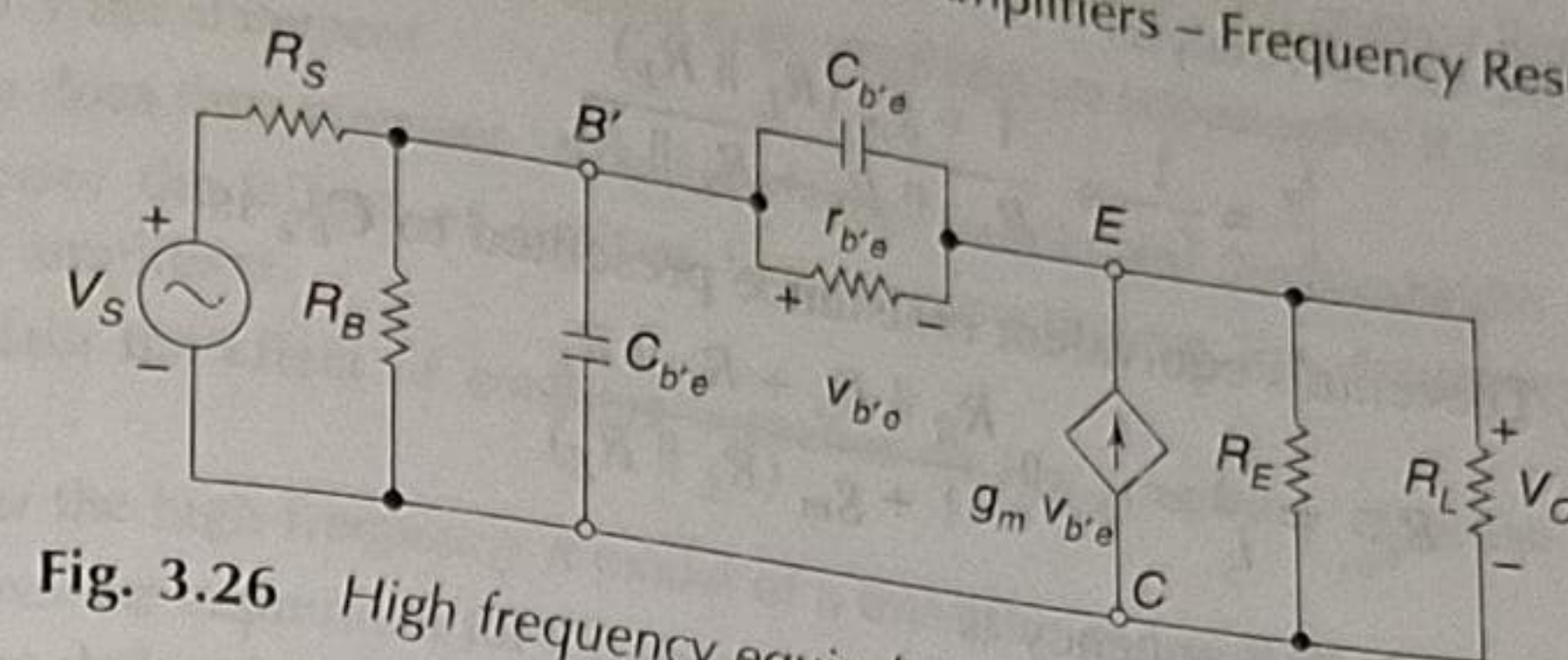
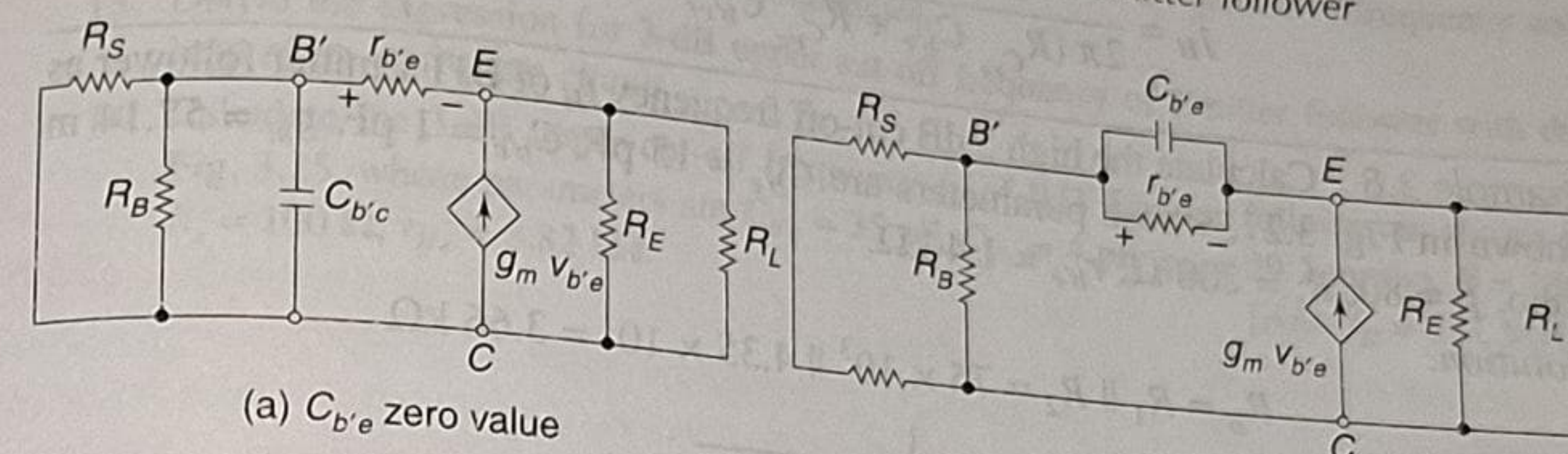
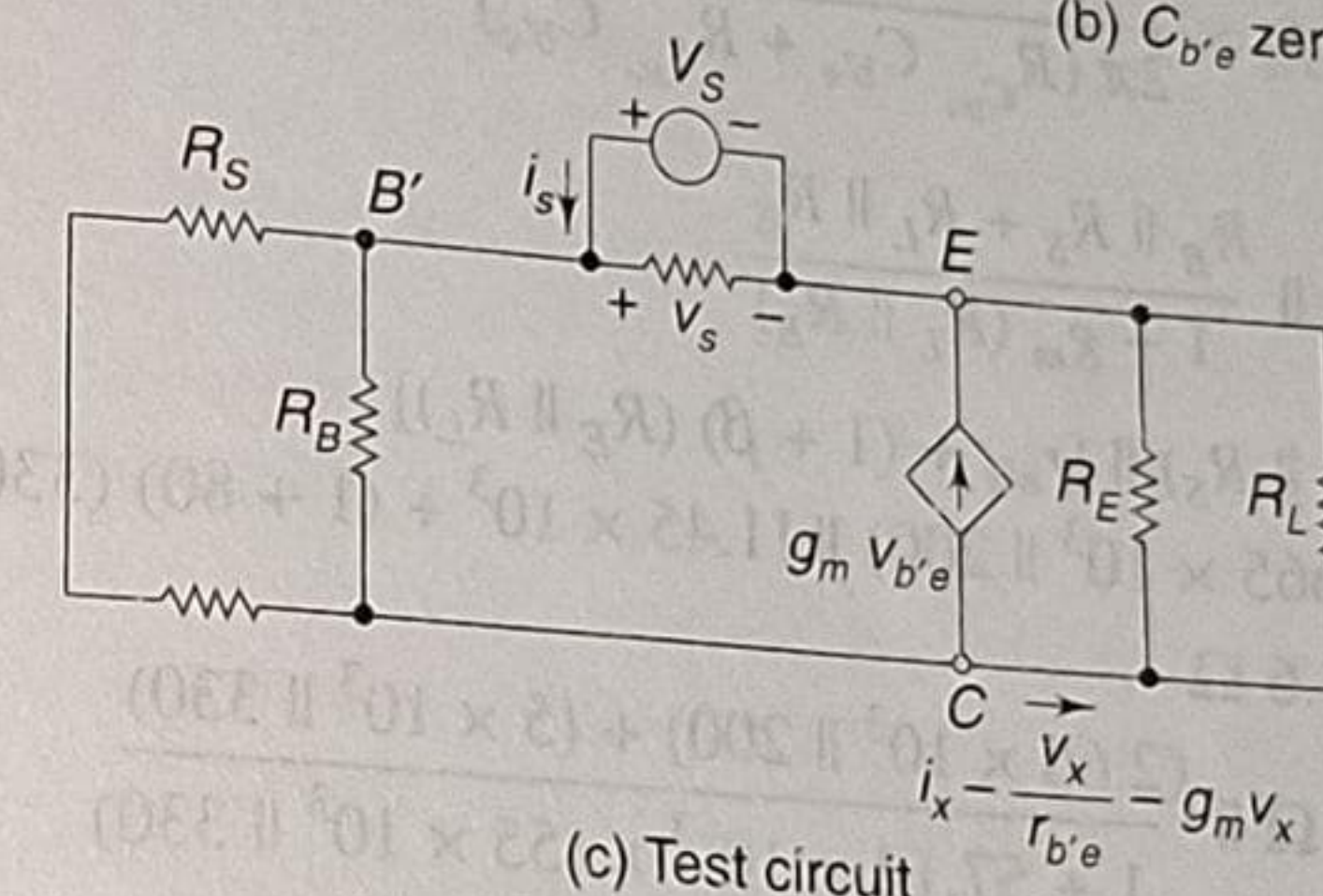


Fig. 3.26 High frequency equivalent circuit of emitter follower



(a) $C_{b'e}$ zero value

(b) $C_{b'e}$ zero value



(c) Test circuit

Fig. 3.27 High frequency equivalent circuits of emitter follower

Looking the resistance to the right of $C_{b'c}$, we get

$$R_{C_{b'e2}} = [r_{b'e} + (1 + \beta) (R_E \parallel R_L)]$$

Thus, Thevenin's equivalent resistance presented to $C_{b'c}$ is

$$R_{C_{b'c}} = R_{C_{b'e1}} \parallel R_{C_{b'e2}} = (R_B \parallel R_S) \parallel [r_{b'e} + (1 + \beta) (R_E \parallel R_L)]$$

Assuming that $C_{b'e}$ is open-circuited, its equivalent circuit will be as shown in Fig. 3.27(b). To determine $R_{C_{b'e}}$, we can remove $C_{b'e}$ and apply a test voltage v_x as shown in Fig. 3.27(c).

Using Kirchhoff's Voltage law around the loop formed by R_B in parallel R_S and by R_L in parallel with R_E , we get

$$v_x = (R_B \parallel R_S) \left(i_x - \frac{v_x}{r_{b'e}} \right) + (R_L \parallel R_E) \left(i_x - \frac{v_x}{r_{b'e}} - g_m v_x \right)$$

Simplifying, we get

$$i_x (R_B \parallel R_S + R_L \parallel R_E) = v_x \left[1 + \frac{R_B \parallel R_S}{r_{b'e}} + \frac{R_L \parallel R_E}{r_{b'e}} + g_m (R_L \parallel R_E) \right]$$

$$\text{i.e. } \frac{i_x}{v_x} = \frac{1}{r_{b'e}} + \frac{1 + g_m (R_L \parallel R_E)}{R_{b'e} \parallel R_S + R_L \parallel R_E}$$

Therefore, Thevenin's equivalent resistance presented to $C_{b'e}$ is

$$R_{C_{b'e}} = \frac{v_x}{i_x} = r_{b'e} \parallel \frac{R_B \parallel R_S + R_L \parallel R_E}{1 + g_m (R_L \parallel R_E)}$$

Hence, the high 3 dB frequency is

$$f_H = \frac{1}{2\pi (R_{C_{b'e}} C_{b'e} + R_{C_{b'c}} C_{b'c})}$$

Example 3.8 Calculate the high 3 dB cut-off frequency f_H of BJT Emitter follower as shown in Fig. 3.27, whose parameters are $C_{b'e} = 15$ pF, $C_{b'c} = 1$ pF, $g_m = 57.14$ mho, $\beta = 80$, $R_S = 200 \Omega$, $r_{b'e} = 1.4$ k Ω .

Solution:

$$R_B = R_1 \parallel R_2 = 75 \times 10^3 \parallel 4.35 \times 10^3 = 3.66 \text{ k}\Omega$$

$$\text{We know that } f_H = \frac{1}{2\pi (R_{C_{b'e}} C_{b'e} + R_{C_{b'c}} C_{b'c})}$$

$$\text{where } R_{C_{b'e}} = r_{b'e} \parallel \frac{R_B \parallel R_S + R_L \parallel R_E}{1 + g_m (R_L \parallel R_E)}$$

$$R_{C_{b'e}} = (R_B \parallel R_S) \parallel [r_{b'e} + (1 + \beta)(R_E \parallel R_L)]$$

$$R_{C_{b'e}} = (2.665 \times 10^3 \parallel 200) \parallel [1.45 \times 10^3 + (1 + 80)(330 \parallel 55 \times 10^3)]$$

$$= 184.6 \Omega$$

$$R_{C_{b'c}} = 1.4 \Omega \parallel \frac{(2.66 \times 10^3 \parallel 200) + (5 \times 10^3 \parallel 330)}{1 + 57.14 \times 10^{-3} \times (55 \times 10^3 \parallel 330)}$$

$$= 26.02 \Omega$$

Hence, the 3 dB upper cut-off frequency is

$$f_H = \frac{1}{2\pi (R_{C_{b'e}} C_{b'e} + R_{C_{b'c}} C_{b'c})}$$

$$= \frac{1}{2\pi (20.02 \times 15 \times 10^{-12} + 184.6 \times 1 \times 10^{-12})}$$

$$= 276.9 \text{ MHz}$$

Review Questions

1. What is the significance of logarithmic scale?
2. Define "bel" and "Decibel".
3. What is dBm?
4. How is half power bandwidth calculated?
5. Explain in detail about logarithms and decibels.
6. What are half-power frequencies?

7. "The cut-off frequencies of single stage amplifiers are influenced by R-C combinations"—Justify the statement.
8. How does time constant ' τ ' and rise time ' t_r ' influence the Bandwidth of amplifiers.
9. Discuss the effect of Emitter bypass capacitor on low frequency response of BJT amplifiers.
10. Explain the effect of coupling capacitor on low frequency response of BJT amplifiers.
11. Draw the high frequency π model of a transistor and explain it.
12. Derive the expression for CE short-circuit current gain as a function of frequency and hence define the α and β cut-off frequencies.
13. Derive the expression for 3-dB upper cut off frequency of Emitter follower with the help of its equivalent circuit.
14. Calculate the 3 dB upper cut-off frequency f_H of BJT Emitter follower as shown in Fig. 3.25, whose parameters are $C_{b'e} = 35$ pF, $C_{b'c} = 4$ pF, $g_m = 39.2$ mmho, $\beta = 150$, $R_S = 100 \Omega$, $r_{b'e} = 3.82$ k Ω .
[Ans. $f_H = 281$ MHz]

The application process is a very essential function in most of the analog circuits. As an example, in digital signal processing, (i) when the input signal is large, (ii) when a large amount of noise or a subsequent stage is needed, or (iii) when needed to provide a right logical level of signal to the next circuit. The amplification process plays an important role in the digital systems also.

This chapter deals with the low frequency behaviour of the single stage CMOS amplifiers. Both the large-signal and the small-signal characteristics of such circuit are analyzed. The techniques and models are developed that pave the way for further understanding of more complex systems.

The basic concepts are reviewed, followed by the description of the five types of amplifiers, namely, (i) common source, (ii) common gate, (iii) common drain or source follower, and (iv) push-pull configurations. In each of these topics, a simple model is proposed, followed by the analysis of various phenomena of MOSFETs such as the channel-length modulation and the body effect.

3.2 BASIC CONCEPTS

The input-output characteristic of an amplifier as shown in Fig. 4.1 is normally a nonlinear function. It can be approximated as a univocal over a finite signal range as given by

$$v_o(t) = a_0 + a_1 v_i(t) + a_2 v_i^2(t) + a_3 v_i^3(t) + \dots + a_n v_i^n(t) \quad (3.1)$$

The input and output signals may be voltage or current. Assuming a sinusoidal input signal, $v_i(t) = V_m \sin \omega t$, we can consider

$$v_o(t) = a_0 + a_1 V_m \sin \omega t + a_2 V_m^2 \sin^2 \omega t + \dots$$

Here, a_0 is the operating point or the bias point and a_1 is the small-signal gain. As long as the value of V_m is small, the variation in the bias point is negligible, provided a reasonable approximation, and higher order terms are insignificant. In other words, $\Delta a_0 = 0$, $\Delta a_1 = 0$, $\Delta a_2 = 0$, $\Delta a_3 = 0$, $\Delta a_4 = 0$, $\Delta a_5 = 0$, $\Delta a_6 = 0$, $\Delta a_7 = 0$, $\Delta a_8 = 0$, $\Delta a_9 = 0$, $\Delta a_{10} = 0$, $\Delta a_{11} = 0$, $\Delta a_{12} = 0$, $\Delta a_{13} = 0$, $\Delta a_{14} = 0$, $\Delta a_{15} = 0$, $\Delta a_{16} = 0$, $\Delta a_{17} = 0$, $\Delta a_{18} = 0$, $\Delta a_{19} = 0$, $\Delta a_{20} = 0$, $\Delta a_{21} = 0$, $\Delta a_{22} = 0$, $\Delta a_{23} = 0$, $\Delta a_{24} = 0$, $\Delta a_{25} = 0$, $\Delta a_{26} = 0$, $\Delta a_{27} = 0$, $\Delta a_{28} = 0$, $\Delta a_{29} = 0$, $\Delta a_{30} = 0$, $\Delta a_{31} = 0$, $\Delta a_{32} = 0$, $\Delta a_{33} = 0$, $\Delta a_{34} = 0$, $\Delta a_{35} = 0$, $\Delta a_{36} = 0$, $\Delta a_{37} = 0$, $\Delta a_{38} = 0$, $\Delta a_{39} = 0$, $\Delta a_{40} = 0$, $\Delta a_{41} = 0$, $\Delta a_{42} = 0$, $\Delta a_{43} = 0$, $\Delta a_{44} = 0$, $\Delta a_{45} = 0$, $\Delta a_{46} = 0$, $\Delta a_{47} = 0$, 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